

$T(2) = 2/2 = 1$
 d) $T(2) = 2 \cdot 3 + 1 = 64$ $T(64) = 64/2 = 32$ $T(32) = 32/2 = 16$ $T(16) = 16/2 = 8$
 $T(8) = 8/2 = 4$ $T(4) = 4/2 = 2$ $T(2) = 2/2 = 1$

42. Verify the $3n+1$ conjecture for these integers.

a) 16 b) 11 c) 35 d) 113
Soln: a) $T(16) = 16/2 = 8$ $T(8) = 8/2 = 4$ $T(4) = 4/2 = 2$ $T(2) = 2/2 = 1$
 b) $T(11) = 11 \cdot 3 + 1 = 34$ $T(34) = 34/2 = 17$ $T(17) = 17 \cdot 3 + 1 = 52$ $T(52) = 52/2 = 26$ $T(26) = 26/2 = 13$ $T(13) = 13 \cdot 3 + 1 = 40$ $T(40) = 40/2 = 20$ $T(20) = 20/2 = 10$ $T(10) = 10/2 = 5$ $T(5) = 5 \cdot 3 + 1 = 16$ $T(16) = 16/2 = 8$ $T(8) = 8/2 = 4$ $T(4) = 4/2 = 2$ $T(2) = 2/2 = 1$
 c) $T(35) = 35 \cdot 3 + 1 = 106$ $T(106) = 106/2 = 53$ $T(53) = 3 \cdot 53 + 1 = 160$ $T(160) = 160/2 = 80$ $T(80) = 80/2 = 40$ $T(40) = 40/2 = 20$ $T(20) = 20/2 = 10$ $T(10) = 10/2 = 5$ $T(5) = 3 \cdot 5 + 1 = 16$ $T(16) = 16/2 = 8$ $T(8) = 8/2 = 4$ $T(4) = 4/2 = 2$ $T(2) = 2/2 = 1$
 d) $T(113) = 3 \cdot 113 + 1 = 340$ $T(340) = 340/2 = 170$ $T(170) = 170/2 = 85$ $T(85) = 3 \cdot 85 + 1 = 256$ $T(256) = 256/2 = 128$ $T(128) = 128/2 = 64$ $T(64) = 64/2 = 32$ $T(32) = 32/2 = 16$ $T(16) = 16/2 = 8$ $T(8) = 8/2 = 4$ $T(4) = 4/2 = 2$ $T(2) = 2/2 = 1$

43. Prove or disprove that you can use dominoes to tile the standard checkerboard with two adjacent corners removed (that is, corners that are not opposite)

Solution: WLOG, Let the white square of row 1 column 1 and the black square of row 8 column 1. Since a standard 8×8 checkerboard has 32 black and 32 white squares each, we are now

X	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W